10. Scenario: You are working on a data visualization project and need to create basic plots using Matplotlib. You have a dataset containing the monthly sales data for a company, including the month and corresponding sales values. Your task is to develop a Python program that generates line plots and bar plots to visualize the sales data.
Question:

1. How would you develop a Python program to create a line plot of the monthly sales data?

2. How would you develop a Python program to create a bar plot of the monthly sales data?

(A)

AIM:

To create a line plot showing the monthly sales performance of a company using Matplotlib.

PROCEDURE:

1. Import the Matplotlib library.
2. Define the months and corresponding sales values.
3. Use plt.plot() to create a line plot.
4. Add a title, axis labels, and grid.
5. Display the plot using plt.show().

PROGRAM CODE:

```
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']

sales = [10000, 12000, 15000, 13000, 18000, 20000]
```

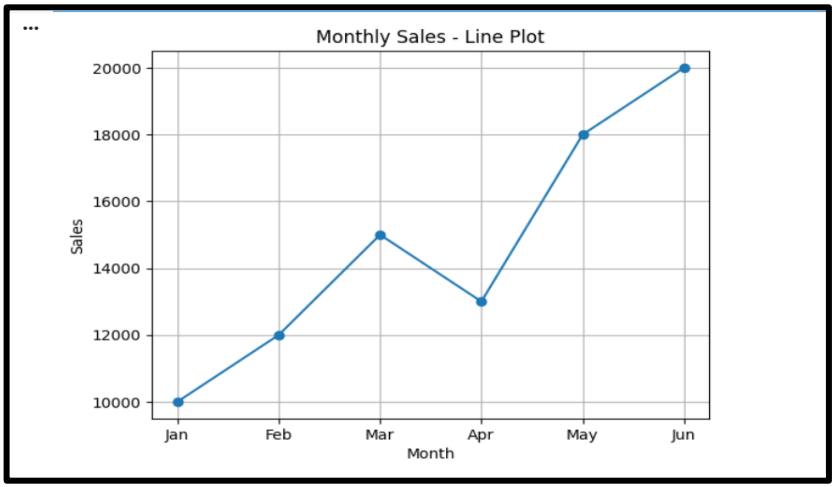
```
plt.plot(months, sales, marker='o')
plt.title('Monthly Sales - Line Plot')
plt.xlabel('Month')
plt.ylabel('Sales')
plt.grid(True)
plt.show()
```

INPUT:

Months = Jan, Feb, Mar, Apr, May, Jun

Sales = 10000, 12000, 15000, 13000, 18000, 20000

OUTPUT:



RESULT:

Thus, a line plot was successfully created to visualize monthly sales trends using Matplotlib.

(B)

AIM:

To create a bar plot to compare the monthly sales of a company using Matplotlib.

PROCEDURE:

1. Import the Matplotlib library.
2. Define the months and sales values.
3. Use plt.bar() to create a bar plot.
4. Add a title and axis labels.
5. Display the plot using plt.show().

PROGRAM CODE:

```
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']

sales = [10000, 12000, 15000, 13000, 18000, 20000]

plt.bar(months, sales)

plt.title('Monthly Sales - Bar Plot')

plt.xlabel('Month')

plt.ylabel('Sales')

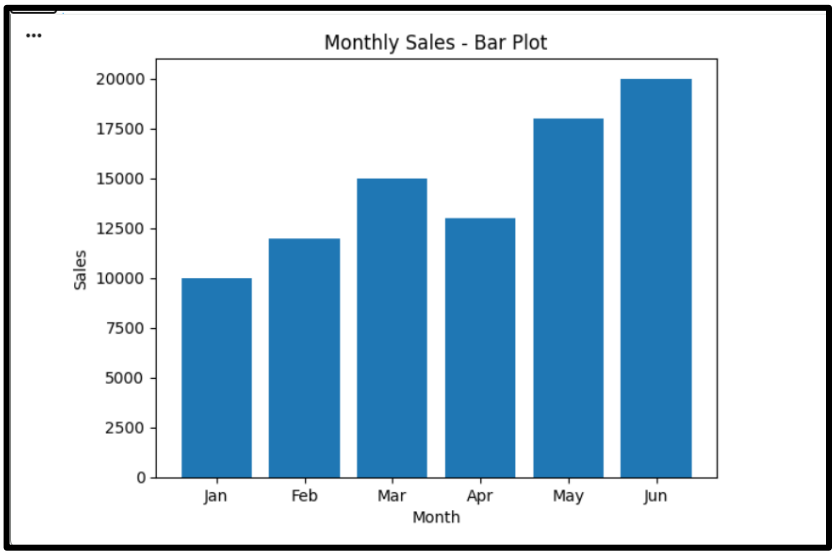
plt.show()
```

INPUT:

Months = Jan, Feb, Mar, Apr, May, Jun

Sales = 10000, 12000, 15000, 13000, 18000, 20000

OUTPUT:



RESULT:

Thus, a bar plot was successfully created to compare monthly sales values using Matplotlib.